Europe PMC Funders Group
Author Manuscript
N Engl J Med. Author manuscript; available in PMC 2026 January 15.

Published in final edited form as:
N Engl J Med. 2026 January 15; 394(3): 232–242. doi:10.1056/NEJMoa2501396.

Vision Restoration with the PRIMA System in Geographic Atrophy Due to AMD

Frank G. Holz, M.D.[1], **Yannick Le Mer, M.D.**[2], **Mahiul M.K Muqit, M.D. PhD**[3], **Lars-Olof Hattenbach, M.D. PhD**[4], **Andrea Cusumano, M.D. PhD**[5], **Salvatore Grisanti, M.D.**[6], **Laurent Kodjikian, M.D. PhD**[7], **Marco Andrea Pileri, M.D.**[8], **Frederic Matonti, M.D. PhD**[9], **Eric Souied, M.D. PhD**[10], **Boris V. Stanzel, M.D. PhD**[11], **Peter Szurman, M.D. PhD**[12], **Michel Weber, M.D. PhD**[13], **Karl Ulrich Bartz-Schmidt, M.D.**[14], **Nicole Eter, M.D.**[15], **Marie Noelle Delyfer, M.D. PhD**[16], **Jean François Girmens, M.D.**[17], **Koen A. van Overdam, M.D. PhD**[18], **Armin Wolf, M.D. PhD**[19], **Ralf Hornig, PhD**[20], **Martina Corazzol, PhD**[20], **Frank Brodie, M.D.**[21], **Lisa Olmos de Koo, M.D. MBA**[22], **Daniel Palanker, PhD**[23,*], **José-Alain Sahel, M.D.**[2,17,24,*]

[1]Department of Ophthalmology, University of Bonn, Germany.

[2]Hôpital Fondation A. de Rothschild, Paris, France

[3]Vitreoretinal Service, Moorfields Eye Hospital, London, Institute of Ophthalmology, University College London, both in the United Kingdom.

[4]Department of Ophthalmology, Ludwigshafen Academic Teaching Hospital, Ludwigshafen am Rhein, Germany

[5]Department of Experimental Medicine, University of Rome Tor Vergata, Italy.

[6]Department of Ophthalmology, Medical Center Schleswig-Holstein, University of Luebeck, Germany.

[7]Service d'Ophtalmologie, Hôpital Universitaire de la Croix-Rousse, Hospices Civils de Lyon, Laboratoire MATEIS, UMR-CNRS 5510, INSA Université Claude Bernard Lyon 1, both in France.

[8]Azienda Ospedaliera San Giovanni Addolorata, Rome, Italy.

[9]Centre Monticelli Paradis, Marseille, L'Université d'Aix-Marseille, CNRS, INT, Inst Neuroscience Timone, Marseille, both in France.

[10]Department of Ophthalmology, Intercommunal Hospital of Créteil and Henri Mondor Hospital, France

[11]Eye Clinic Sulzbach, Knappschaft Hospital Saar, Department of Ophthalmology, University of Bonn, both in Germany.

[12]Eye Clinic Sulzbach, Knappschaft Hospital Saar

This work is licensed under a BY 4.0 International license.

Correspondence to: Frank G. Holz.

Corresponding author: Frank G. Holz, Department of Ophthalmology, University of Bonn, Venusberg-Campus 1 | D-53127 Bonn | Germany, Frank.Holz@ukbonn.de .
*co-senior authors

Disclosures
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

Europe PMC Funders Author Manuscripts
Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

[13]Nantes University, Nantes University Hospital, Department of Ophthalmology, France.

[14]University Eye Hospital Tübingen, Germany

[15]Department of Ophthalmology, University of Münster Medical Center, Germany.

[16]Bordeaux University Hospital, France.

[17]Hôpital National des 15-20, Inserm-DGOS CIC1423, Paris, France.

[18]Department of Ophthalmology, Erasmus Medical Center, Rotterdam, Rotterdam Ophthalmic Institute, Rotterdam, , both in the Netherlands.

[19]Department of Ophthalmology. University of Ulm, Germany.

[20]Science Corporation, Alameda, United States of America

[21]Department of Ophthalmology, University of California San Francisco, San Francisco, United States of America.

[22]Department of Ophthalmology, University of Washington, Seattle, United States of America.

[23]Department of Ophthalmology, Stanford University, Stanford, United States of America.

[24]Department of Ophthalmology, University of Pittsburgh School of Medicine, Pittsburgh, United States of America, , Sorbonne Université, Paris, France.

Abstract

Background: Geographic atrophy (GA) due to age-related macular degeneration (AMD) is the leading cause of irreversible blindness and affects over 5 million people globally. Currently, no therapies exist to restore vision to the affected persons. The PRIMA system combines a subretinal photovoltaic implant and near-infrared light-projecting glasses to restore sight to areas of central retinal atrophy.

Methods: In an open-label, baseline-controlled, multicenter, prospective, single-arm clinical study, the vision of participants with GA and a visual acuity of logMAR ≥1.2 was assessed with and without PRIMA glasses at six and twelve months. The primary endpoints were the proportion of participants with improvement in visual acuity (VA) of logMAR 0.2 or more and the number and severity of study-related serious adverse events at 12 months.

Results: Thirty-two of thirty-eight enrolled participants reached the 12-month endpoint. Of the six who did not, three died, one withdrew, and two were unavailable for testing. Twenty-six (81.3%; 95% CI: 63.56–92.79%) had an improved prosthetic VA of at least logMAR 0.2 versus baseline ($p<0.001$) and in 19 participants, there were 26 serious adverse events. Twenty-one of these events (81%) occurred within the first two months, of which 20 (95%) were resolved within two months of implantation. There was no significant change in peripheral natural VA compared to baseline.

Conclusions: In this study of a limited number of participants, the PRIMA system restored central vision with significant improvements of VA in participants with GA due to AMD. (Funded by Science Corporation, and others; Trial Reg: NCT04676854)

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Background

Age-related macular degeneration (AMD) is the most common cause of incurable blindness in the elderly.[1,2] Geographic atrophy (GA), an advanced stage of dry AMD, leads to progressive, irreversible death of photoreceptors in the outer retina, causing profound vision loss.[3] Globally, GA affects approximately 5 million people and in North America it is responsible for ~20% of all cases of legal blindness.[2–4]

The first therapies for GA, designed to slow disease progression using complement inhibitors (pegcetacoplan and avacincaptad pegol), were recently approved in the United States, and require monthly or bimonthly intravitreal injections.[5–7] However, no approved therapies, investigational approaches, or cell therapies, have shown meaningful visual improvement.[8–12] In a healthy retina, photoreceptors transduce light into electrical signals which are then processed by the inner retina and transmitted to the brain. Due to a loss of photoreceptors in GA, light is not transduced into electrical signals, leading to an absolute scotoma.[3] The PRIMA neurostimulation system (hereinafter, PRIMA system) replaces the lost photoreceptors with a photovoltaic implant.[13]

The PRIMA implant is a 2×2mm wide and 30μm thick crystalline silicon array comprising 378 photovoltaic pixels, each 100μm in size.[14,15] It is implanted subretinally within the atrophic lesion. A frame-mounted camera on the PRIMA glasses captures images, and projects them, after processing onto the implant, using near-infrared (880nm) light (Figure 1). The implant's pixels convert near-infrared light into electric pulses to stimulate retinal bipolar cells, restoring the flow of visual information.[14] Unlike a wired prosthesis, the photovoltaic nature of the implant enables wireless operation combined with a straightforward implantation technique.[16,17] The lens in the PRIMA glasses is transparent, so participants can perceive natural and prosthetic vision simultaneously.[18]

After extensive pre-clinical testing,[18–20] a first-in-human clinical trial evaluated the feasibility of the PRIMA system in five participants with GA.[21] Although the primary endpoint of the feasibility trial was to elicit prosthetic perception, after optimization and training, three participants reliably recognized sequences of letters and had acuity closely matching the 20/420 maximum resolution allowed by the pixel size (100 μm).[15] At 4 years, these patients could read small fonts, with a mean visual acuity (VA) of 20/135.[21] The PRIMAvera study was conducted to assess safety and efficacy of the PRIMA system.

Methods

Study Design

PRIMAvera is an open-label, baseline-controlled, non-randomized, multicenter, prospective, single-arm, confirmatory clinical study involving 38 participants with GA due to advanced AMD (NCT04676854). The study, conducted at 17 clinical sites across 5 European countries, evaluated the efficacy and safety of the PRIMA system in participants with GA in both eyes and an atrophic patch including the fovea with profound central vision loss in at least one eye. PRIMAvera was designed by Science Corporation with support from its advisory board and approved by local ethics committees. Clinical authors conducted

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

the study, Science Corporation analyzed the data, and the Data Safety and Monitoring Board (DSMB) reviewed it. The corresponding and co-senior authors wrote the manuscript, together with authors from Science Corporation. All authors reviewed and approved the final manuscript for submission. The authors vouch for the accuracy and completeness of data and data analyses, along with the conduct of the trial according to the protocol (available at NEJM.org).

Daniel Palanker and Jose Alain Sahel co-senior authors vouch for the data and the analysis.

Inclusion criteria included: age 60 years or older with a confirmed diagnosis of GA due to AMD in both eyes validated by fundus autofluorescence imaging; VA of logMAR 1.2 (Snellen equivalent 20/320) or worse in the study eye; fovea-involving atrophy greater than the implant size (>2.4mm in diameter) in the study eye.

Study Endpoints

The primary efficacy endpoint was the proportion of participants (lower bound >50%) achieving a clinically meaningful visual acuity improvement, defined as logMAR ≥0.2 (≥10 letters) on a standard ETDRS chart at 12 months.[15,21–24] Secondary efficacy endpoints included the proportion of participants with an improvement of VA at 6 months, mean VA improvement at 6 and 12 months, impairment as assessed by Impact-of-Vision-Impairment (IVI) questionnaire at 6 and 12 months, and central visual perception at 12 months.[25,26]

The primary safety endpoint was the number and severity of device- and procedure-related serious adverse events (SAEs) at 12 months. Secondary safety endpoints were the number and severity of all related adverse events (AEs; serious and non-serious), the change in best corrected natural VA (without the PRIMA glasses) compared to baseline, and the percentage of compliant implantations at 4 weeks post-operation. Additional follow-up is planned for up to 36 months.

Visual Function Assessment

We tested the best corrected VA of the study eye with and without the PRIMA glasses 6 and 12 months after surgery; participants could adjust brightness and zoom level at will. After 12 months, participants answered a survey on home use of PRIMA and their ability to perform various visual tasks.

Statistical Analysis

Data from all 38 participants were included in the primary analysis. Missing data for primary efficacy endpoints were simulated using multiple imputation models (see Supplementary Appendix) and an additional analysis based on observed data was performed. A binomial test evaluated the proportion of participants achieving an improvement of logMAR ≥0.2, compared to a predefined success threshold of 50%. For the primary endpoint, confidence intervals for binomial testing were computed with the exact method for observed data and Logit method for multiple imputation.

Mean VA improvement from baseline to 12 months was assessed as a secondary endpoint. A sequential gatekeeping approach was tested against a null hypothesis of mean improvement

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

of zero as well as of logMAR 0.2, based on a one-sided α=0.025. For secondary endpoints, exact confidence intervals were used. VA data were analyzed under three conditions: 1) without PRIMA glasses, 2) with PRIMA glasses, and 3) participant's choice, to reflect home use. The safety cohort analysis was based on observed data without imputation and included all device- and procedure-related SAEs (defined by ISO14155). The DSMB reviewed all device- and procedure-related AEs. All statistical analyses were performed using SAS version 9.4. Surgical procedure, vision training, and additional study methods are detailed in the Supplementary Appendix.

Results

The PRIMA system was implanted in 38 participants (18 male/20 female) with a mean age of 78.9 years (±6.41 SD) (Table S4). This group is broadly representative of the global GA population (Table S5). Figure 2 shows a representative pre- and post-implantation set of fundus and optical coherence tomography (OCT) images. At 12 months, 32 participants were available for evaluation (Figure S1). Six participants were not assessed due to death (3), withdrawal (1) or unavailability for testing (2) (Table S6).

Prosthetic vision

Of 32 participants, 30 demonstrated central perception at 12 months. Prosthetic VA (without zoom and image processing) was measured using Landolt C optotypes directly displayed on the PRIMA glasses, without use of the camera (Table S7). The mean VA was logMAR 1.32±0.16 (Snellen 20/417). This average value matches the sampling limit of the 100 μm pixels used by PRIMA (logMAR 1.32, see Supplementary Appendix).

Efficacy of the system

At 12 months, 81.3% (95% CI: 63.56 to 92.79%; 26/32; p<0.001) of observed participants demonstrated a clinically meaningful improvement in VA (logMAR ≥0.2). Using multiple imputation with pre-specified covariates to account for missing data from six participants, an estimated 79.9% of all participants (95% CI: 65.58 to 94.16%; p<0.001) would have achieved meaningful improvement.

At 12 months, participants showed mean VA improvements under both conditions: with PRIMA glasses (logMAR 0.49; 95% CI: 0.35 to 0.63; range: -0.32 to 1.18) and with participant's choice (logMAR 0.51; 95% CI: 0.39 to 0.64; range: -0.04 to 1.18), corresponding to improvements of 24.5 and 25.5 letters, respectively. With participant's choice, those who used natural vision (N=4) had no meaningful VA improvement (logMAR -0.01; range: -0.04 to 0.02). The VA improvement, with predefined thresholds of logMAR 0 and 0.2, was found to be statistically significant using a sequential t-test (logMAR 0: p<0.001; logMAR 0.2: p<0.001) (Table S9). The best improvement was logMAR 1.18 (59 letters; Figure 3). At 6 months, 20 out of 35 participants had an improvement of logMAR 0.2. The mean improvement was higher from baseline with the PRIMA system (logMAR 0.32; 95% CI: 0.16 to 0.47) and with participant's choice (logMAR 0.38; 95% CI: 0.25 to 0.51) (Figure 3). In contrast, when participants did not use PRIMA glasses, there was no change in VA from baseline (6 months: logMAR 0.01; 95% CI: -0.04 to 0.06; 12 months:

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

logMAR 0.00; 95% CI: -0.05 to 0.04). Participants dynamically adjusted zoom settings (range 1×–12×) and brightness during VA testing (see Table S8 for maximum zoom usage). Single-participant VA trajectories and further information on VA outcomes based on years since AMD diagnosis prior to implantation can be found in Figure S2 and Tables S10 and S11.

Digital enhancement features including contrast, brightness, and zoom applied to the camera feed helped participants to see a wide range of visual stimuli. At 12 months, 84.4% of participants (27/32) could read letters, numbers and words using prosthetic vision at home (see supplementary videos S1–S3); 68% (22/32) reported medium-to-high user satisfaction with the PRIMA system. However, results from the IVI survey did not indicate behavioral changes (Table S12).

Post-hoc analysis

After the primary analysis, a similar analysis was performed for an improvement of logMAR ≥0.3 and found 78.1% of participants demonstrated an improvement at 12 months (see Table 1).

Safety

Twenty-six of device- and procedure-related SAEs occurred in 19 unique participants (Table 2) in 12 months. All SAEs were related either to the implantation procedure alone or a combination of the procedure and device. No SAE was related to the device alone. Of the 26 SAEs, 21 (81%) occurred within 2 months of surgery, of which 20 resolved within 2 months of occurrence. Twenty-two (85%) were classified as mild or moderate. Four (15%) were severe (macular hole, ocular hypertension, retinal detachment, proliferative vitreoretinopathy). The most frequent SAE was ocular hypertension (6 events; 23%), which occurred between 1 day and 3 weeks post-implantation, and all cases have been resolved. Five participants had peripheral retinal breaks; all were treated intraoperatively, and none resulted in rhegmatogenous retinal detachment. Three participants developed a subretinal hemorrhage during implantation, which stopped spontaneously, or after increasing infusion pressure. One had recurrent subretinal hemorrhage associated with choroidal neovascularization at 9 months. This participant and another, who also had choroidal neovascularization, were treated with intravitreal anti-vascular endothelial growth factor (VEGF) therapy. Three SAEs were full-thickness macular holes; two of these required moving the implant away from the hole to improve the functionality of the device. The choroidal fold, retinal detachment and proliferative vitreoretinopathy occurred in the same participant, but were successfully treated by surgery and silicone oil tamponade.

As demonstrated in Figure 2 and assessed by imaging (OCT, fundus photography), the inner retinal structure in the atrophic area does not appear to have been altered by the subretinal placement of the implanted chip.

The average area of atrophy in the study eyes increased from baseline to 12 months by 8.5 mm^2, compared with 2.5 mm^2 increase in the fellow eyes; this was attributed to post-surgical changes. There was no change in mean natural VA compared to baseline in the study eyes (Figure 3). Of the 32 patients at 12 months, 11 of the 14 participants affected by

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

SAEs (79%) and 15 of the 18 participants unaffected by SAEs (83%) showed improvement of logMAR ≥0.2 with the PRIMA system at 12 months. All study-related SAEs were prespecified in the risk analysis, and none were life-threatening. On review of these and other data obtained in this study, the DSMB concluded that the PRIMA system's benefits outweigh the risks of implantation.

Discussion

The PRIMA system demonstrated meaningful visual improvement in participants with profound vision loss due to fovea-involving GA from AMD, in a representative population sample.[27] This study demonstrated that 81% of participants at 12 months gained 10 or more letters (logMAR ≥0.2), and 78% of participants gained 15 or more letters (logMAR ≥0.3), with the PRIMA system. The mean improvement at 12 months was 25.5 letters (logMAR 0.51), and the maximum was 59 letters (logMAR 1.18). Moreover, 84% of participants reported using the device at home for reading letters, numbers, or words. They were able to read fonts smaller (up to 20/42 Snellen) than the sampling limit of 100 μm pixels (about 20/400 Snellen) using digital enhancements like zoom. Other low-vision aids like extraocular magnifiers or implantable telescopes enlarge images to enable utilization of the retina beyond the edges of atrophy[24,28], achieves a typical average gain of logMAR 0.24[24] and magnify the full visual field. In contrast, PRIMA restores vision to the area of scotoma and only magnifies the central prosthetic vision, leaving natural peripheral vision unaffected.[14] While the native resolution of PRIMA approximates that of the peripheral retina in the participants (~logMAR 1.3), the additional abilities to zoom and enhance contrast and make other image improvements allowed participants to gain acuity beyond this level.

Prosthetic vision was previously attempted with the epiretinal implant Argus II,[16,17] the subretinal implant Alpha IMS,[29] and a 44 channel suprachoroidal implant.[30] ARGUS II had limited resolution due to large electrode pitch (highest reported visual acuity was logMAR 1.8)[31] and stimulation of axons from remote retinal ganglion cells, resulting in a distorted retinotopic map.[32,33] Suprachoroidal implants are relatively far from the retinal neurons, and therefore provide comparatively low resolution in the range of logMAR 3.0.[30]

PRIMA allows the use of eye movement, as opposed to head-scanning,[23] to reorient the camera. Eye-scanning improves resolution beyond the sampling limit,[34–36] presumably for similar reasons as super-resolution algorithms implemented with conventional cameras.[37] Unlike epiretinal implants, PRIMA stimulates bipolar cells rather than ganglia, which preserves features of inner retinal signal processing.[18,38]

Our data, collected over 12 months, demonstrate that PRIMA can be safely implanted under the atrophic macula, restoring central vision while preserving residual natural peripheral vision. Our previous feasibility study demonstrated stable anatomy with minimal reduced thickness of the inner retina over 36 months and VA logMAR 0.64 (32 letters) at 48 months.[21,39] The wireless design simplifies implantation and reduces the surgical and post-operative risks compared to wired retinal implants with permanent openings of the eye

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

globe.[31] PRIMA is thin (about half the height of a photoreceptor) and integrates with the retina without mechanical fixation, such as retinal tacks used with other implants.[24]

The complications we observed in this study are consistent with the risks associated with vitrectomy and subretinal surgery, namely sub-retinal hemorrhage and CNV.[40,41] Two participants (5.2%) developed CNV, which was successfully treated with intravitreal anti-VEGF therapy on a *pro re nata* regimen. CNV can occur in GA even without any intervention[5] and has also been observed in patients following retinal surgical procedures. CNV incidence has been reported in vitrectomized eyes with dry AMD as 16% after 2.6 years.[42]

The implantation of a subretinal prosthesis may benefit from advanced intraoperative imaging technologies. Increased atrophy in the study eye frequently mirrored the areas of retinal bleb formation and retinotomy. Notably, no significant change in native vision was observed. The study was not powered to detect a difference in the IVI questionnaire and the questionnaire may be insufficiently sensitive for our participants, who have extremely low vision.

The 12-month results of this pivotal clinical trial demonstrated that the PRIMA subretinal implant can restore meaningful central vision in GA due to AMD, enabling visual tasks, such as reading and writing. While no implants have been explanted in humans, the wireless design allows for replacements with higher resolution next-generation chips,[43,44] or multiple modules to “tile” the atrophic area with minimal incision.[45]

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

The authors thank the DSMB members Sophie J. Bakri, Sebastian Wolf, and Thomas J. Wolfensberger for their careful review of all safety data. We also thank Pr. Ecosse Lamoureux and Pr. Eva Fenwick for their expertise in Impact of Vision Impairment analysis and their valuable guidance. In addition to the clinical institutions affiliated with the authors, we also acknowledge Association de Réadaptation et de Réinsertion pour l’Autonomie des Déficients Visuels (Marseille, France), CLER Basse Vision Domaine de S antifontaine (Nancy, France), Formation Insertion et Réadaptation pour Déficients visuels (Lyon, France), and the Independent Regional High Specialization Center for Visual Rehabilitation at Tor Vergata University (Rome, Italy) for participation in rehabilitation.

Funding Statement

The PRIMAvera study was financed by Science Corporation, Alameda, CA, (previously Pixium Vision SA, Paris, France). The site in London was partially supported by the National Institute for Health and Care Research Biomedical Research Centre based at Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology.

References

1. Wong WL, Su X, Li X, et al. Global prevalence of age-related macular degeneration and disease burden projection for 2020 and 2040: a systematic review and meta-analysis. Lancet Glob Health. 2014; 2 (2) e106–16. [PubMed: 25104651]

Europe PMC Funders Author Manuscripts

2. Rudnicka AR, Jarrar Z, Wormald R, Cook DG, Fletcher A, Owen CG. Age and Gender Variations in Age-related Macular Degeneration Prevalence in Populations of European Ancestry: A Meta-analysis. Ophthalmology. 2012; 119 (3) 571–80. [PubMed: 22176800]

3. Fleckenstein M, Mitchell P, Freund KB, et al. The Progression of Geographic Atrophy Secondary to Age-Related Macular Degeneration. Ophthalmology. 2018; 125 (3) 369–90. [PubMed: 29110945]

4. Holz FG, Strauss EC, Schmitz-Valckenberg S, Van Lookeren Campagne M. Geographic Atrophy. Ophthalmology. 2014; 121 (5) 1079–91. [PubMed: 24433969]

5. Heier JS, Lad EM, Holz FG, et al. Pegcetacoplan for the treatment of geographic atrophy secondary to age-related macular degeneration (OAKS and DERBY): two multicentre, randomised, double-masked, sham-controlled, phase 3 trials. The Lancet. 2023; 402 (10411) 1434–48. [PubMed: 37865470]

6. Patel SS, Lally DR, Hsu J, et al. Avacincaptad pegol for geographic atrophy secondary to age-related macular degeneration: 18-month findings from the GATHER1 trial. Eye Lond Engl. 2023; 37 (17) 3551–7. DOI: 10.1038/s41433-023-02497-w [PubMed: 36964259]

7. Kang C. Avacincaptad Pegol: First Approval. Drugs. 2023; 83 (15) 1447–53. [PubMed: 37814173]

8. Girgis S, Lee LR. Treatment of dry age-related macular degeneration: A review. Clin Experiment Ophthalmol. 2023; 51 (8) 835–52. [PubMed: 37737509]

9. Holz FG, Sadda SR, Staurenghi G, et al. Imaging Protocols in Clinical Studies in Advanced Age-Related Macular Degeneration. Ophthalmology. 2017; 124 (4) 464–78. [PubMed: 28109563]

10. Deng Y, Qiao L, et al. Age-related macular degeneration: Epidemiology, genetics, pathophysiology, diagnosis, and targeted therapy. Genes Dis. 2022; 9 (1) 62–79. DOI: 10.1016/j.gendis.2021.02.009 [PubMed: 35005108]

11. Humayun MS, Clegg DO, Dayan MS, et al. Long-term Follow-up of a Phase 1/2a Clinical Trial of a Stem Cell-Derived Bioengineered Retinal Pigment Epithelium Implant for Geographic Atrophy. Ophthalmology. 2024; 131 (6) 682–91. [PubMed: 38160882]

12. Loewenstein A, Trivizki O. Future perspectives for treating patients with geographic atrophy. Graefes Arch Clin Exp Ophthalmol. 2023; 261 (6) 1525–31. DOI: 10.1007/s00417-022-05931-z [PubMed: 36520185]

13. Lorach H, Goetz G, Smith R, et al. Photovoltaic restoration of sight with high visual acuity. Nat Med. 2015; 21 (5) 476–82. DOI: 10.1038/nm.3851 [PubMed: 25915832]

14. Palanker D, Le Mer Y, Mohand-Said S, Sahel JA. Simultaneous perception of prosthetic and natural vision in AMD patients. Nat Commun. 2022; 13 (1) 513. doi: 10.1038/s41467-022-28125-x [PubMed: 35082313]

15. Palanker D, Le Mer Y, Mohand-Said S, Muqit M, Sahel JA. Photovoltaic Restoration of Central Vision in Atrophic Age-Related Macular Degeneration. Ophthalmology. 2020; 127 (8) 1097–104. DOI: 10.1016/j.ophtha.2020.02.024 [PubMed: 32249038]

16. Da Cruz L, Dorn JD, Humayun MS, et al. Five-Year Safety and Performance Results from the Argus II Retinal Prosthesis System Clinical Trial. Ophthalmology. 2016; 123 (10) 2248–54. DOI: 10.1016/j.ophtha.2016.06.049 [PubMed: 27453256]

17. Schaffrath K, Schellhase H, Walter P, et al. One-Year Safety and Performance Assessment of the Argus II Retinal Prosthesis: A Postapproval Study. JAMA Ophthalmol. 2019; 137 (8) 896. doi: 10.1001/jamaophthalmol.2019.1476 [PubMed: 31145440]

18. Ho E, Lei X, Flores T, et al. Characteristics of prosthetic vision in rats with subretinal flat and pillar electrode arrays. J Neural Eng. 2019; 16 (6) 066027 doi: 10.1088/1741-2552/ab34b3 [PubMed: 31341094]

19. Arens-Arad T, Farah N, Lender R, et al. Cortical Interactions between Prosthetic and Natural Vision. Curr Biol. 2020; 30 (1) 176–182. e2 doi: 10.1016/j.cub.2019.11.028 [PubMed: 31883811]

20. Ho E, Smith R, Goetz G, et al. Spatiotemporal characteristics of retinal response to network-mediated photovoltaic stimulation. J Neurophysiol. 2018; 119 (2) 389–400. DOI: 10.1152/jn.00872.2016 [PubMed: 29046428]

21. Muqit MMK, Le Mer Y, Olmos De Koo L, Holz FG, Sahel JA, Palanker D. Prosthetic Visual Acuity with the PRIMA Subretinal Microchip in Patients with Atrophic Age-Related Macular Degeneration at 4 Years Follow-up. Ophthalmol Sci. 2024; 4 (5) 100510 doi: 10.1016/j.xops.2024.100510 [PubMed: 38881600]

Europe PMC Funders Author Manuscripts

22. Dziedziak J, Kasarełło K, Cudnoch-Jędrzejewska A. Dietary Antioxidants in Age-Related Macular Degeneration and Glaucoma. Antioxidants. 2021; 10 (11) 1743. doi: 10.3390/antiox10111743 [PubMed: 34829613]
23. Anderson AG, Ratnam K, Roorda A, Olshausen BA. High-acuity vision from retinal image motion. J Vis. 2020; 20 (7) 34. doi: 10.1167/jov.20.7.34 [PubMed: 32735342]
24. Boyer D, Freund KB, Regillo C, Levy MH, Garg S. Long-term (60-month) results for the implantable miniature telescope: efficacy and safety outcomes stratified by age in patients with end-stage age-related macular degeneration. Clin Ophthalmol Auckl NZ. 2015; 9: 1099–107. DOI: 10.2147/OPTH.S86208 [PubMed: 26124633]
25. Lamoureux EL, Pallant JF, Pesudovs K, Hassell JB, Keeffe JE. The Impact of Vision Impairment Questionnaire: an evaluation of its measurement properties using Rasch analysis. Invest Ophthalmol Vis Sci. 2006; 47 (11) 4732–41. [PubMed: 17065481]
26. Lamoureux EL, Pallant JF, Pesudovs K, Rees G, Hassell JB, Keeffe JE. The impact of vision impairment questionnaire: an assessment of its domain structure using confirmatory factor analysis and rasch analysis. Invest Ophthalmol Vis Sci. 2007; 48 (3) 1001–6. [PubMed: 17325138]
27. Rahimy E, Khan MA, Ho AC, et al. Progression of Geographic Atrophy: Retrospective Analysis of Patients from the IRIS® Registry (Intelligent Research in Sight). Ophthalmol Sci. 2023; 3 (4) 100318 doi: 10.1016/j.xops.2023.100318 [PubMed: 37274013]
28. eSight - Electronic eyewear for the visually impaired. ESight Eyewear; [Internet] Available from: https://www.esighteyewear.com/ [cited 2024 Dec 18]
29. Stingl K, Schippert R, Bartz-Schmidt KU, et al. Interim Results of a Multicenter Trial with the New Electronic Subretinal Implant Alpha AMS in 15 Patients Blind from Inherited Retinal Degenerations. Front Neurosci. 2017; 11: 445. doi: 10.3389/fnins.2017.00445 [PubMed: 28878616]
30. Petoe MA, Abbott CJ, Titchener SA, et al. A Second-Generation (44-Channel) Suprachoroidal Retinal Prosthesis: A Single-Arm Clinical Trial of Feasibility. Ophthalmol Sci. 2025; 5 (1) 100525 doi: 10.1016/j.xops.2024.100525 [PubMed: 39328823]
31. Humayun MS, Dorn JD, da Cruz L, et al. Interim results from the international trial of Second Sight's visual prosthesis. Ophthalmology. 2012; 119 (4) 779–88. DOI: 10.1016/j.ophtha.2011.09.028 [PubMed: 22244176]
32. Nanduri D, Fine I, Horsager A, et al. Frequency and amplitude modulation have different effects on the percepts elicited by retinal stimulation. Invest Ophthalmol Vis Sci. 2012; 53 (1) 205–14. DOI: 10.1167/iovs.11-8401 [PubMed: 22110084]
33. Behrend MR, Ahuja AK, Humayun MS, Chow RH, Weiland JD. Resolution of the epiretinal prosthesis is not limited by electrode size. IEEE Trans Neural Syst Rehabil Eng Publ IEEE Eng Med Biol Soc. 2011; 19 (4) 436–42. DOI: 10.1109/TNSRE.2011.2140132 [PubMed: 21511569]
34. Abraham C, Farah N, Gerbi-Zarfati L, Harpaz Y, Zalvesky Z, Mandel Y. Active photonic sensing for super-resolved reading performance in simulated prosthetic vision. Biomed Opt Express. 2019; 10 (3) 1081–96. DOI: 10.1364/BOE.10.001081 [PubMed: 30891331]
35. Ratnam K, Harmening W, Roorda A. Fixational eye movements improve visual performance at the sampling limit. J Vis. 2015; 15 (12) 1272.
36. Ratnam K, Domdei N, Harmening WM, Roorda A. Benefits of retinal image motion at the limits of spatial vision. J Vis. 2017; 17 (1) 30. doi: 10.1167/17.1.30 [PubMed: 28129414]
37. Ur H, Gross D. Improved resolution from subpixel shifted pictures. CVGIP Graph Models Image Process. 1992; 54 (2) 181–6.
38. Ho E, Lorach H, Goetz G, et al. Temporal structure in spiking patterns of ganglion cells defines perceptual thresholds in rodents with subretinal prosthesis. Sci Rep. 2018; 8 (1) 3145 doi: 10.1038/s41598-018-21447-1 [PubMed: 29453455]
39. Muqit MMK, Mer YL, Holz FG, Sahel JA. Long-term observations of macular thickness after subretinal implantation of a photovoltaic prosthesis in patients with atrophic age-related macular degeneration. J Neural Eng. 2022; 19 (5) doi: 10.1088/1741-2552/ac9645 [PubMed: 36174540]
40. Fujiwara N, Tomita G, Yagi F. Incidence and Risk Factors of Iatrogenic Retinal Breaks: 20-Gauge versus 25-Gauge Vitrectomy for Idiopathic Macular Hole Repair. J Ophthalmol. 2020; 2020 5085180 doi: 10.1155/2020/5085180 [PubMed: 32104593]

Europe PMC Funders Author Manuscripts

41. Kaufmann GT, Gupta O, Yu J, et al. Vitreoretinal outcomes following secondary intraocular lens implantation with pars plana vitrectomy. Retina Phila Pa. 2024; 44 (8) 1337–43. [PubMed: 39047127]

42. Obeid A, Ali FS, Deaner JD, Gao X, Hsu J, Chiang A. Outcomes of Pars Plana Vitrectomy for Epiretinal Membrane in Eyes With Coexisting Dry Age-related Macular Degeneration. Ophthalmol Retina. 2018; 2 (8) 765–70. [PubMed: 31047527]

43. Bhuckory MB, Monkongpitukkul N, Shin A, et al. Enhancing Prosthetic Vision by Upgrade of a Subretinal Photovoltaic Implant in situ. 2024; cited 2024 Dec 15 doi: 10.1038/s41467-025-58084-y [PubMed: 40118873]

44. Wang B-Y, Chen ZC, Bhuckory M, et al. Electronic photoreceptors enable prosthetic visual acuity matching the natural resolution in rats. Nat Commun. 2022; 13 (1) 6627. doi: 10.1038/s41467-022-34353-y [PubMed: 36333326]

45. Lee DY, Lorach H, Huie P, Palanker D. Implantation of Modular Photovoltaic Subretinal Prosthesis. Ophthalmic Surg Lasers Imaging Retina. 2016; 47 (2) 171–4. DOI: 10.3928/23258160-20160126-11 [PubMed: 26878451]

Europe PMC Funders Author Manuscripts

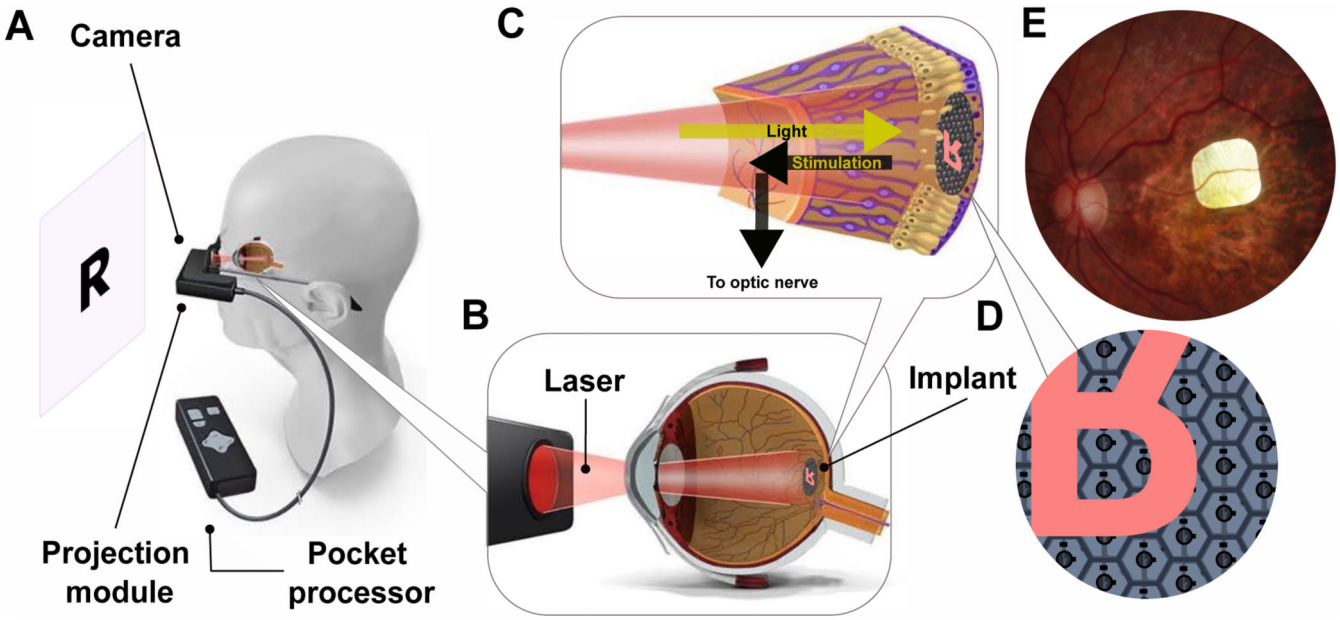


Figure 1. Rendering of the PRIMA system components.

(A) The system includes the PRIMA glasses with a camera and projection module, a pocket processor, and the implant. The captured image (e.g., a letter R) is processed and projected onto the implant by NIR (880nm) light. (B) Projection of NIR light onto the implant. (C) The implant, activated by the NIR light, stimulates the remaining nerve cells, which process the information in the retina and then transfer it via the optic nerve to the brain. (D) Photodiodes in each pixel of the subretinal implant convert light into electric currents to stimulate the nearby inner retinal neurons. To utilize flicker fusion for stable perception, the projector operates at a 30Hz frame rate, and perceptual brightness is adjusted by controlling the pulse duration from 0.7 to 9.8ms at peak irradiance of 3.5mW/mm^2. Each pixel comprises the active electrode in the center, a hexagonal return electrode mesh, and two photodiodes filling the space between the active and the return electrode. (E) Fundus photograph of a participant's retina with the PRIMA implant. The implant is 2×2mm in size, composed of 100μm pixels.

Europe PMC Funders Author Manuscripts
Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

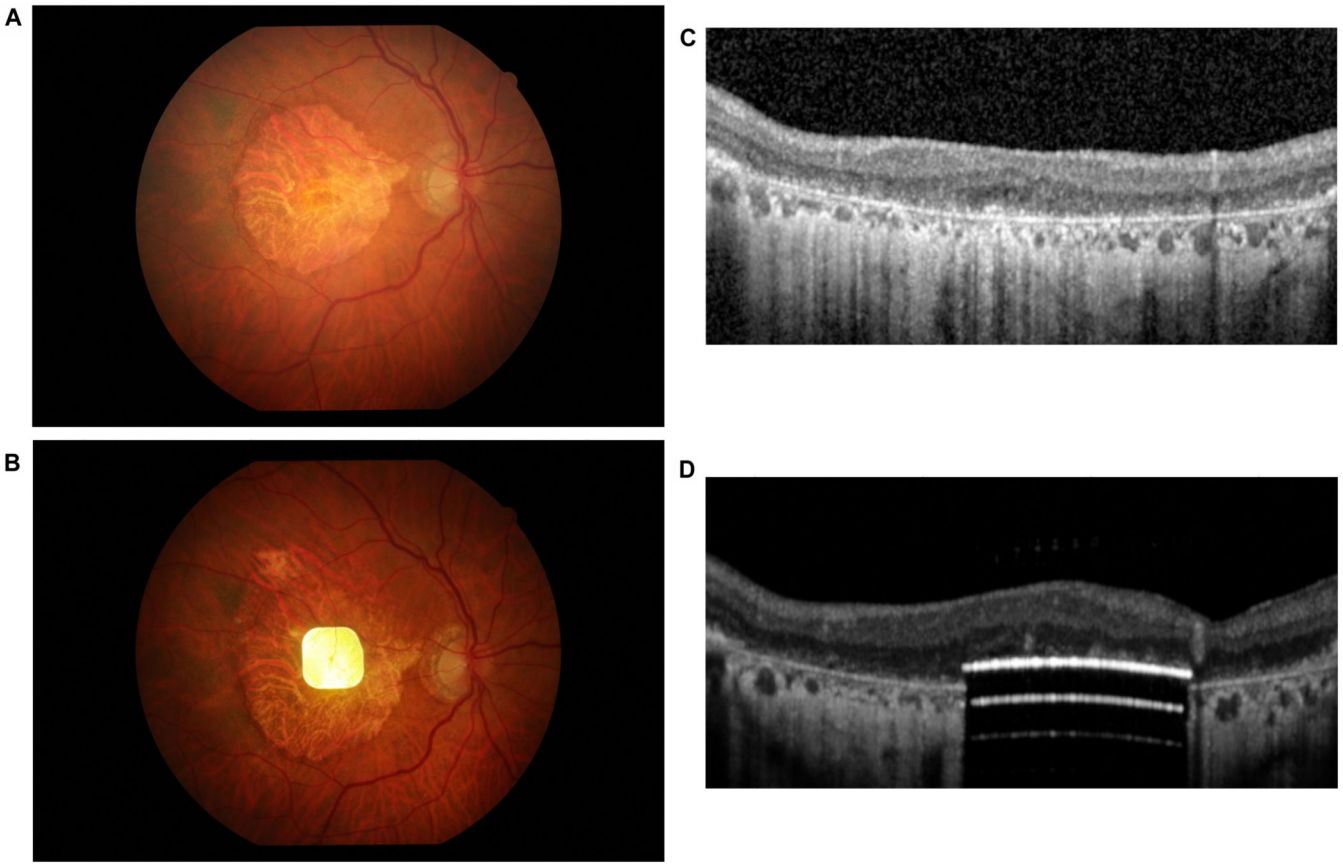


Figure 2. Subretinal implant in a study participant's eye.

This participant had an atrophy size of 21.49 mm^2 at baseline and was implanted five years after AMD diagnosis. (A) Color fundus photograph demonstrates GA prior to implantation. (B) The same eye with a subretinal implant at 12 months post-implantation. The green arrows indicate the direction of OCT scanning. (C) OCT image of the eye pre-implantation. (D) OCT image demonstrating the implant under the degenerated retina in proximity to the inner nuclear layer.

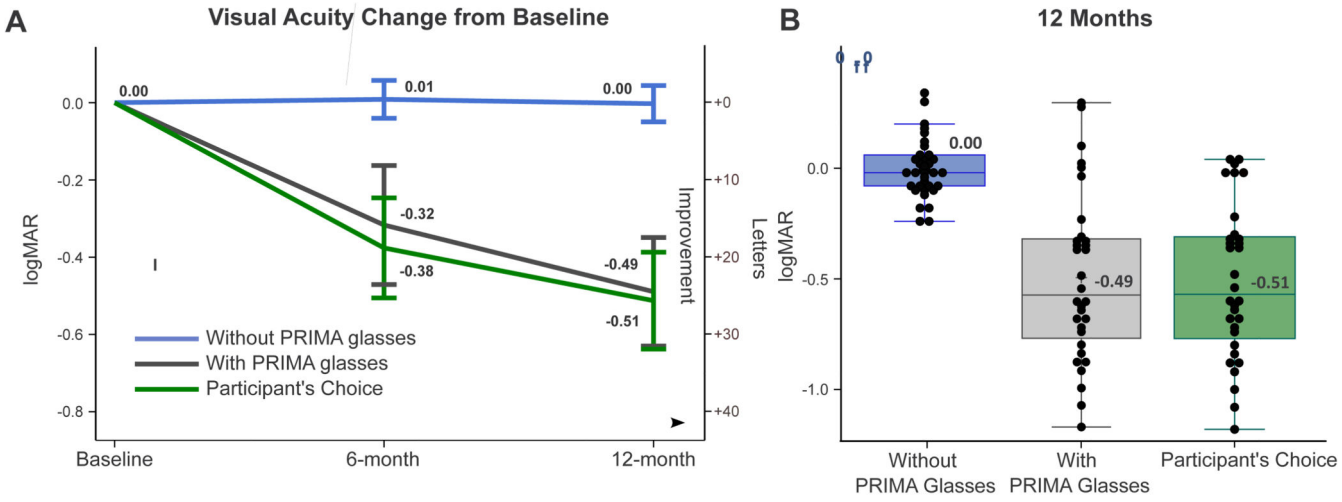


Figure 3. Change of VA from baseline (ETDRS).

(A) Progression of VA over time, measured at baseline, 6 months, and 12 months under three conditions: without PRIMA glasses (blue line), with PRIMA glasses (grey line), and participant's choice (green line). Change in VA is expressed in both logMAR (primary y-axis) and letters (secondary y-axis). Data points represent the mean, with error bars indicating the 95% confidence intervals. Negative logMAR values indicate improvement in VA. (B) Box plots depicting logMAR change from baseline at 12 months without PRIMA glasses, with PRIMA glasses and participant's choice. Individual participants' data are represented by black dots.

Europe PMC Funders Author Manuscripts
Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Table 1 Primary efficacy endpoint: VA improvement of logMAR ≥0.2 and ≥0.3 at 12 months.

Group	Improvement logMAR ≥ 0.2		Improvement logMAR ≥ 0.3[c]		Participants
	n/N and % of Participants	95% CI p-value[‡]	n/N and % of Participants	95% CI	(Total; Imputed)
Primary Endpoint (Multiple Imputation)[a]	NA[†]/38 79.90%	65.58–94.16% <.001	NA[†]/38 77.20%	62.18–92.14%	38; 6
Observed Data[b]	**26/32** 81.30%	63.56–92.79% <.001	**25/32** 78.10%	60.03–90.72%	32; 0

Binomial test results of two different scenarios.

[a]Primary Endpoint (Multiple Imputation): Multiple Imputation analysis was performed on missing data.

[b]Observed Data: Analysis of observed data on implanted participants with a non-missing 12-month follow-up result.

[c]Post-hoc analysis.

[†]The multiple imputation results do not provide the exact proportion of participants achieving the primary efficacy endpoint but rather an estimated percentage, computed over 100 imputed data sets.

[‡]The p value is given for the null hypothesis of proportion of participants - with an improvement of logMAR 0.2 or more - less than or equal to 50%.

Exact intervals were used when analyzing observed data while Logit intervals were used for multiple imputation.

Europe PMC Funders Author Manuscripts

Europe PMC Funders Author Manuscripts

Table 2 Device- or procedure-related SAEs at the 12-month follow-up.

SAE Type	N participants with SAE	% participants with SAE	Number of events
Ocular hypertension	6	15.79%	6
Peripheral retinal break	5	13.16%	5
Macular hole	3	7.89%	3
Subretinal hemorrhage	3	7.89%	4
Choroidal neovascularization	2	5.26%	2
Choroidal hemorrhage	1	2.63%	1
Choroidal fold	1	2.63%	1
Proliferative vitreoretinopathy	1	2.63%	1
Retinal detachment	1	2.63%	1
Retinal hemorrhage	1	2.63%	1
Thrombophlebitis	1	2.63%	1
Total	**25**		**26**